THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of factors of 16 is: 1

(a) 7 **(b)** 6 **(c)** 4 **(d)** 5

2. The converse of the proposition "If a number is divisible by 12, then it is divisible by 4" is: 1

(a) If it is not true that a number is divisible by 12, then it is not true that it is divisible by 4.

(b) If a number is divisible by 12, then it is divisible by 4.

(c) If it is not true that it is divisible by 4, then a number is divisible by 12.

(d) If a number is divisible by 4, then it is divisible by 12.

3. Which of the following numbers is divisible by 9? 1

(a) 810 **(b)** 812 **(c)** 811 **(d)** 815

4. The proposition "If a parallelogram is a rectangle, then its diagonals are equal" is true. Its converse "If its diagonals are equal, then a parallelogram is a rectangle" is: 1

(a) It cannot be decided **(b)** True only for some values

(c) True **(d)** False

5. In "If an integer n is even, then n^2 is even", the hypothesis is: 1

(a) an integer n is even, then n^2 is even

(b) an integer n is even

(c) it is not true that an integer n is even

(d) n^2 is even

6. In "If an integer n is odd, then n^2 is odd", the conclusion is: 1

(a) an integer n is odd

(b) n^2 is odd

(c) it is not true that n^2 is odd

(d) an integer n is odd, then n^2 is odd

7. The statement "The sum of two prime numbers is even" is: 1

(a) false **(b)** true

(c) true only for prime numbers **(d)** true only for even numbers

8. "Every multiple of 6 is even" can be written in if–then form as: 1

(a) If not (a number is a multiple of 6), then it is even.

(b) If a number is a multiple of 6, then it is even.

(c) A number is a multiple of 6 or it is even.

(d) If it is even, then a number is a multiple of 6.

9. The number of factors of 20 is: 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 5

10. The converse of the proposition "If a number is a perfect square, then it has an odd number of factors" is: 1

(a) If it is not true that it has an odd number of factors, then a number is a perfect square.
(b) If it is not true that a number is a perfect square, then it is not true that it has an odd number of factors.
(c) If a number is a perfect square, then it has an odd number of factors.
(d) If a number has an odd number of factors, then it is a perfect square.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the number of factors of 25. Is 25 a perfect square? 2

12. Write the converse of "If an integer n is even, then n^2 is even". Is it true? 2

13. Write "Even numbers are divisible by 2" in if–then form and name the hypothesis. 2

14. Find the number of factors of 18. Is 18 a perfect square? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If two angles are vertically opposite, then the two angles are equal". Is the converse true? Justify. 3

16. Show that the statement "The sum of two irrational numbers is irrational" is false by giving a counterexample. 3

17. Is a triangle with sides 9 cm, 12 cm and 16 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Write the converse of "If a number is divisible by 9, then the sum of its digits is divisible by 9". Is the converse true? Justify. 3

18. Show that the statement "The difference of two primes is prime" is false by giving a counterexample. 3

OR

Is a triangle with sides 20 cm, 21 cm and 29 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Verify with 17 and 9 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 17 and 18. 5

20. Verify with 23 and 5 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 22 and 23. 5

END OF QUESTION PAPER — **Total: 40 Marks**